Reliable In-Car Voice Assistance with Target Speaker Extraction

Full system report, benchmark record, ablations, latency, and Jetson transfer package

Car-TSE research project

14 October 2026

Evidence status. This is a research-engineering report, not a production sign-off. The project has zero consented real-driver sessions and no matching six-input capture device. All positive end-to-end numbers are synthetic or cross-corpus proxies; the negative streams are useful diagnostic exposures but are not a substitute for a speaker-consistent vehicle study. The current readiness label is `research_engineering_preview` and `production_ready=false`.

Contents

1 Executive summary **3**
- 1.1 Goal . . . 3
- 1.2 System . . . 3
- 1.3 Headline results . . . 3
- 1.4 What is strong . . . 4
- 1.5 What is weak or not proven . . . 4

2 Research question and intended contribution **4**
- 2.1 Research question . . . 4
- 2.2 Contribution boundary . . . 4

3 Architecture **5**
- 3.1 Runtime data flow . . . 5
- 3.2 Runtime checkpoints . . . 5

4 Datasets and data contracts **5**
- 4.1 Training and evaluation data . . . 5
- 4.2 Licensing and redistribution . . . 6
- 4.3 Required real-driver study . . . 6

5 Training methodology **6**
- 5.1 Stage 1: target-speaker extraction . . . 6
- 5.2 Stage 2: target presence . . . 8
- 5.3 Stage 3: wake detection . . . 8
- 5.4 Stage 4: intent . . . 8
- 5.5 Numerical and training safeguards . . . 8

6 Per-stage accuracy results **8**
6.1 Stage 1 target-speaker extraction . . . 8
6.2 Stage 2 presence gate . . . 9
6.3 Stage 3 wake-word models . . . 9
6.4 Stage 3b speaker verification . . . 10
6.5 Stage 4 intent classification . . . 10

7 End-to-end results **11**
7.1 Positive cross-corpus proxies . . . 11
7.2 Long target-absent cabin negative stream . . . 11
7.3 Open-set non-Marvin speech . . . 11
7.4 15-minute continuous streams . . . 12

8 Production readiness and release gates **13**

9 Reproducibility **13**
9.1 Primary artifacts . . . 13
9.2 Representative commands . . . 14
9.3 Jetson validation procedure . . . 14

10 Paper claims that are currently safe **14**

11 Claims that are not currently safe **15**

12 Recommended next experiments **15**

13 Conclusion **15**

A Metric definitions **16**

B Artifact and model transfer inventory **17**

C References and citation placeholders **17**

D Ablation studies **18**
D.1 Component ablation under passenger speech . . . 18
D.2 Temporal and enrollment ablation . . . 18
D.3 Channel and speaker gate ablation . . . 18
D.4 Verifier scale and fusion . . . 19

E Latency, footprint, and deployment **19**
E.1 Measurement contract . . . 19
E.2 Component latency . . . 19
E.3 Composite latency . . . 20
E.4 Streaming runner . . . 20
E.5 ONNX deployment path . . . 20
E.6 Footprint . . . 21

1 Executive summary

1.1 Goal

The project asks whether an in-car assistant can reliably hear a driver, recognize the wake word "Marvin", understand the command that follows, and avoid acting on passenger or background speech. The intended operating conditions include four cabin microphones, vehicle noise, overlapping passenger speech, optional playback reference audio, short speaker enrollment, and a constrained edge deployment.

1.2 System

The implemented cascade is:

1. four-channel cabin audio and an optional aligned playback reference;
2. a learned cabin front end and a 16 kHz SpEx+-style target-speaker extractor;
3. a framewise target-presence gate;
4. a target-conditioned wake detector for "Marvin";
5. a streaming state machine that arms command capture only after an accepted wake;
6. a two-second command window sent to a 31-class Fluent Speech Commands (FSC) intent classifier; and
7. a vehicle action and optional spoken reply.

The wake frame is not directly decoded as the command. A timeout prevents the state machine from remaining armed indefinitely. The retained deployment export is an ONNX composition of the main stages; the PyTorch reference remains the authority for research evaluation.

1.3 Headline results

Table 1: Best current component and system evidence. Percentages are percentages, SI-SDRi is in dB, and false-event rates are events per hour.

Part	Result and interpretation
Stage 1 TSE	Dual-domain held-out regression: +3.86 dB AISHELL-5 SI-SDRi and +0.11 dB CAVEMOVE SI-SDRi; a separate 400-row CAVEMOVE protocol gives +10.98 dB SI-SDRi. This is meaningful synthetic-cabin evidence, not a real-driver result.
Presence	Frame F1 90.44%, FAR 0.90%, FRR 5.49% at 0.5. The gate is useful, but its runtime threshold is separately calibrated.
Target KWS	Narrow held-out operating point: F1 97.40%. This is strong isolated KWS evidence, not a continuous false-accept result.
Speaker verifier	Cabin+Smart Eval1 AUC 0.9706; TPR 79.7% at 0.53% FPR. Same-driver recall remains unproven.
Intent	Official FSC test: 99.31% top-1 and 99.89% top-5; this is the strongest individual component.
Positive proxy	128 clean cross-corpus rows: 100.0% wake and 84.4% intent. The proxy is not same-driver evidence.
Negative proxy	Smart Speaker hard-negative KWS without speaker gate: 23.81 full-path false wakes/h after full-context correction; still unsafe.
Real-cabin negative	AISHELL-5, 18 sessions, 3.889 h: zero full-path events/h; exact 95% upper bound 0.95/h. This is target-absent evidence only.
Latency	Retained full path on the development host: GPU FP16 56.99 ms wake / 76.21 ms command mean; CPU FP32 561.15 ms / 679.57 ms, too slow for a 250 ms hop.

1.4 What is strong

- The pipeline is implemented end to end and has a corrected wake-to-command state transition.
- Target extraction improves held-out synthetic cabin mixtures by approximately **+10.98 dB SI-SDRi** in the original CAVEMOVE protocol.
- Clean FSC intent classification is strong at **99.31% top-1**.
- The compact verifier and presence gate reduce false triggers on some long negative streams, but their operating point is not yet frozen.
- The ONNX composition, parity checks, ring-buffer runner, and release manifest provide a reproducible deployment path.

1.5 What is weak or not proven

- There are **zero real-driver sessions**; therefore same-driver enrollment, wake, command, passenger rejection, and action correctness are not measured.
- The permissive clean candidate has an open-set false-wake failure; its historical 1,209.38/h figure used the pre-correction tail-window evaluator and is not comparable to the current full-context result.
- After excluding incomplete zero-padded tails, the hard-negative KWS candidate produces 23.81 full-path false wakes/h on 3.193 h of non-Marvin speech, still far above the provisional target of 0.1/h. The corrected compact-verifier path produces 1.88/h at KWS 0.50 and gate 0.30, and 154.74/h at the more recall-friendly KWS 0.46 point.
- The largest latency cost is Stage 1 on a three-second, non-causal window. Host CPU ONNX streaming does not keep up with a 250 ms hop.
- The full model package is about 425 MB and has about 106.2 million parameters, dominated by the wav2vec2 intent branch.

2 Research question and intended contribution

2.1 Research question

Can target-speaker extraction and speaker-conditioned gating make an in-car wake-word-to-intent cascade reject passenger speech and vehicle background while preserving driver command recall under a short enrollment constraint?

2.2 Contribution boundary

The individual ingredients are established techniques: time-domain target-speaker extraction, learned presence detection, log-mel keyword spotting, speaker verification, and wav2vec2 intent classification. The potential contribution is the system-level integration and an evaluation protocol that measures error propagation from target extraction through wake, intent, and action. That claim becomes publishable only after a real, speaker-consistent cabin corpus and held-out driver test are collected.

The present report should therefore be used to write a reproducible systems paper or a research-prototype paper. It should not claim state of the art, production readiness, or broad real-world reliability.

3 Architecture

3.1 Runtime data flow

The reference sampling rate is 16 kHz. A nominal inference window contains three seconds of four-channel microphone data and an optional three-second playback reference. Enrollment is a short waveform, nominally no longer than three seconds. The extracted target stream is passed to the presence and target-KWS stages. Once the state machine accepts "Marvin", the next target speech is placed into a two-second command window and classified into one of 31 FSC intents.

3.2 Runtime checkpoints

Table 2: Retained runtime identity used for latency and ONNX parity.

Stage	Checkpoint	Role	Parameters
Stage 1	`stage1_spex_dual_domain_rms010_ft1/best.pt`	dual-domain TSE and cabin front end	10,727,881
Stage 2	`stage2_frame_presence_dual_domain_prewake_ft1/best.pt`	frame presence gate	33,729
KWS	`stage3_kws_gsc_marvin_tse_ft2/best.pt`	target wake detector	93,378
Target KWS	`stage3_target_kws_gsc_marvin_hardneg_aishell5_shark_speaker_ft1/best.pt`	hard-negative target wake detector	889,316
Verifier	`compact_speaker_verifier_cabin_spatial_ft1/best.pt`	compact speaker gate	176,579
Intent	`stage4_intent_wav2vec_ft8_resume_tse_domain/best.pt` and `encoder_best.pt`	FSC command classifier	94,782,879
		Total	**106,203,762**

4 Datasets and data contracts

4.1 Training and evaluation data

Table 3: Datasets and their role. The local repository contains manifests and cached derivatives; licenses must be checked again before redistribution.

Dataset	Role	Measured use	Caveat
LibriSpeech / Libri2Mix	clean source speech, speaker IDs, mixtures	Stage 1 pretraining and speaker-verifier pairs	not vehicle audio
CAVEMOVE	four-channel cabin noise and multichannel mixtures	Stage 1 cabin adaptation, continuous proxies, latency inputs	synthetic or proxy composition; not a continuous driver study
AISHELL-5	real four-channel far-field speech and noise	target-absent negative streams and domain diagnostics	Mandarin; target speaker absent in negative tests; not an English positive set
Google Speech Commands	"Marvin" keyword and non-wake speech	KWS positives, non-wake negatives, hard-negative construction	isolated short commands; not continuous cabin data
Fluent Speech Commands	31 English intents and slots	Stage 4 intent training and official test	clean command corpus; no wake word or cabin scene

Dataset	Role	Measured use	Caveat
Smart Speaker Command Dataset	continuous "Hey Alexa" commands	non-Marvin open-set negative benchmark; 2,110 files / 3.193 h	clean single-channel studio clips duplicated to four channels
Vehicle Interior Sound / cabin-noise sources	road, HVAC, engine and cabin beds	augmentation and noise-only stress tests	licensing and coverage vary by source
Custom in-car sessions	same driver enrollment, wake, command, passenger, playback	manifest and validator exist, but session directory is empty	decisive missing evidence

4.2 Licensing and redistribution

The project did not redistribute the commercial Nexdata 531-hour corpus or the access-restricted MagicHub/ASR-CabNois material. The current model package is a model/checkpoint transfer package; it does not contain the raw training corpora. Before a paper release, add the canonical dataset citations, license texts, and any required access statements.

4.3 Required real-driver study

The minimum pilot protocol is:

1. obtain consent from at least one driver and one passenger, preferably several speakers;
2. record a short driver enrollment sample, then a continuous sequence of driver "Marvin" plus command;
3. include passenger "Marvin", passenger speech without wake, no-command intervals, silence, road/HVAC/wind, and playback/echo;
4. capture four cabin microphones and two aligned playback-reference channels on hardware matching the intended deployment path;
5. label wake and command intervals, driver/passenger identity, seat position, noise condition, action expectation, and timestamps; and
6. keep drivers or sessions used for threshold tuning separate from the held-out final test.

The current audit reports zero sessions and no matching six-input device. Therefore no result in this report is a real-driver result.

5 Training methodology

5.1 Stage 1: target-speaker extraction

The extraction model is a 16 kHz SpEx+-style time-domain target-speaker extractor with a learned multichannel cabin front end. It was transferred from clean speaker mixtures and adapted to CAVEMOVE and AISHELL-5. The promoted dual-domain candidate uses RMS conditioning at 0.10 during the training/evaluation contract. The promotion rule required no regression on either held-out domain.

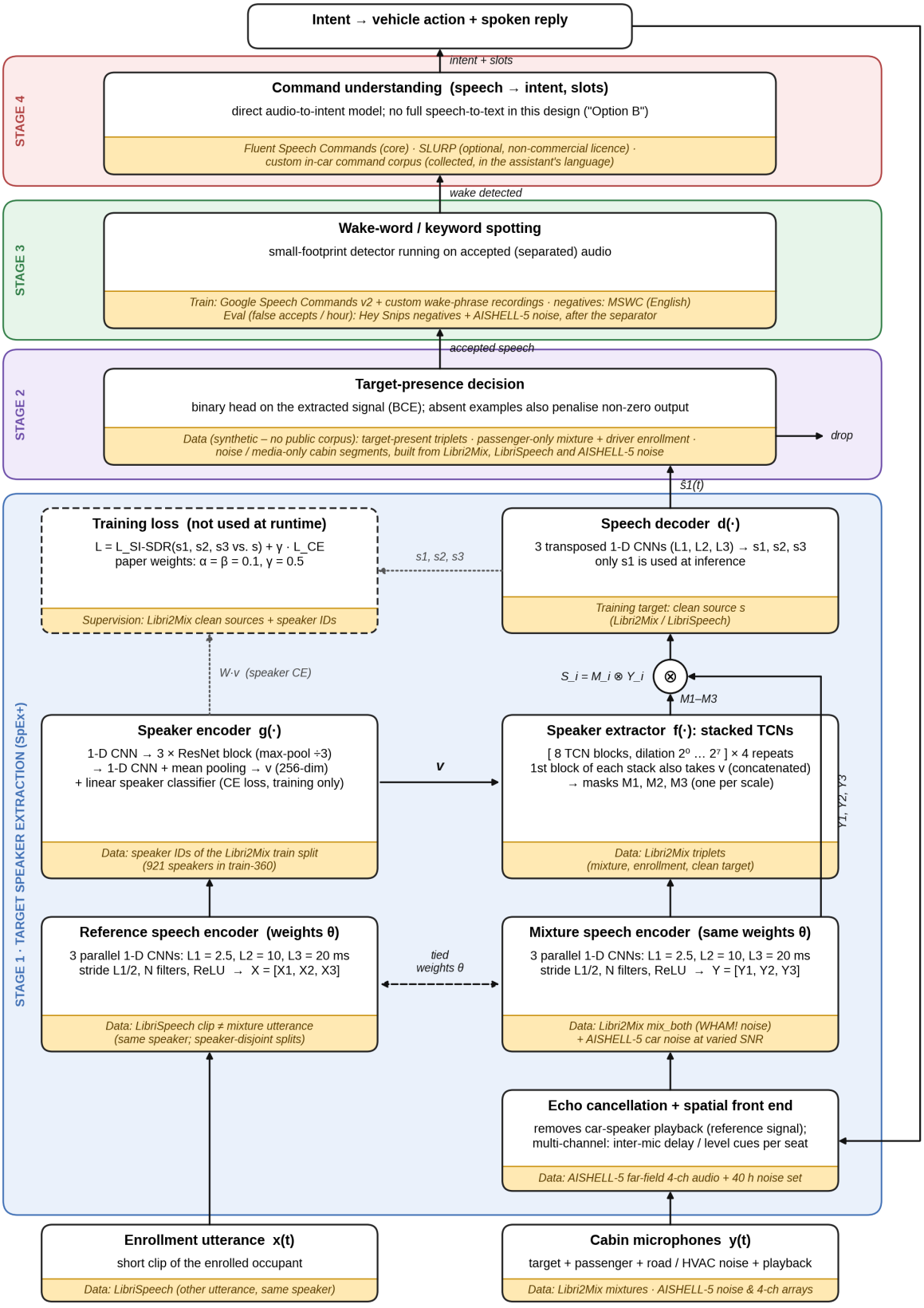


Figure 1: Project architecture and intended production flow. The current runtime implements the learned extraction, presence, wake, state-machine, and intent blocks; real vehicle echo cancellation and seat-zone validation remain open evidence gates.

5.2 Stage 2: target presence

The framewise presence head uses 40 mel bands, a hidden size of 64, a 512-sample FFT, and a 160-sample hop. It is trained on target-present and target-absent mixtures assembled from CAVEMOVE, GSC, AISHELL-5, and hard negatives. The standalone validation threshold is not the runtime threshold: the cascade uses lower wake/command margins and relies on downstream KWS and speaker checks.

5.3 Stage 3: wake detection

The wake word is "Marvin". General KWS uses log-mel features and a compact convolutional classifier. The target-conditioned head additionally sees enrollment and extracted speaker embeddings. Hard-negative variants were trained with AISHELL-5 and Smart Speaker non-wake speech. Many variants were retained under unique checkpoint names; only the selected runtime identity was used for the ONNX release export.

5.4 Stage 4: intent

The intent model uses a wav2vec2-base-960h encoder with a 31-class FSC head. The model is strong on clean FSC but is the dominant resource consumer. A smaller intent family and a causal CRNN were explored, but neither replaces the retained baseline for the main report.

5.5 Numerical and training safeguards

The causal Stage 1 path is opt-in and not promoted. Swapping causal layers into an existing non-causal checkpoint reduced clean validation SI-SDR from 41.67 dB to 4.16 dB. A separate causal fine-tune reported NaN train and validation SI-SDR in its first epoch under mixed precision and was stopped. This result means a streaming model must be trained under the causal contract with finite-gradient guards; an inference-time architectural swap is invalid.

6 Per-stage accuracy results

6.1 Stage 1 target-speaker extraction

Table 4: Dual-domain held-out Stage 1 regression. There are 144 AISHELL-5 and 144 CAVEMOVE validation examples.

Domain	Transfer baseline SI-SDRi	Dual-domain ft1 SI-SDRi	Change
AISHELL-5	2.39 dB	**6.24 dB**	+3.86 dB
CAVEMOVE	10.48 dB	**10.59 dB**	+0.11 dB
Overall	6.43 dB	**8.42 dB**	+1.99 dB

In a separate 400-example CAVEMOVE protocol, the input mixture had mean SI-SDR -2.41 dB and the extracted output had 8.57 dB, giving **+10.98 dB SI-SDRi**. The median improvement was +12.77 dB. These protocols use different splits and should not be merged into one score.

The AISHELL amplitude diagnostic found that fixed input RMS conditioning improved the target-present AISHELL result from approximately -31.26 dB output / +1.47 dB SI-SDRi to -29.01 dB / +4.94 dB at a 0.10 target and -27.76 dB / +6.43 dB at a 0.20 target, while 0.20 cost 0.36 dB on CAVEMOVE. No normalization policy was promoted solely from this small sweep.

6.2 Stage 2 presence gate

Table 5: Frame and chunk views of the dual-domain non-wake validation cache: 1,926 training examples and 482 validation chunks.

View	Accuracy	Precision	Recall	F1	FAR	FRR
Frame, threshold 0.50	98.84%	86.71%	94.51%	90.44%	0.90%	5.49%
Chunk, threshold 0.50	99.79%	99.18%	100.00%	99.59%	0.28%	0.00%
Best frame F1, threshold 0.65	98.98%	90.50%	92.10%	91.29%	0.60%	7.90%

The earlier scalar-gate calibration illustrates why standalone accuracy is not enough: at the default scalar threshold 0.50, the measured FAR was 96.25% with 0% FRR on an imbalanced split; a threshold near 0.90 reduced that diagnostic to approximately 17.5% FAR / 8.9% FRR. The runtime frame head and downstream gates are therefore evaluated at operating points chosen for cascade rejection, not at the standalone F1 threshold.

6.3 Stage 3 wake-word models

Table 6: Representative KWS scores. F1 is split-dependent; continuous false events per hour are the deployment-relevant measure.

Variant	Precision	Recall	F1	Note
Original cabin KWS	52.29%	82.05%	63.87%	narrow cabin distribution
TSE-domain held-out KWS	95.94%	96.92%	96.43%	narrow TSE-domain regression
Selected high-precision TSE-domain point	98.94%	95.90%	**97.40%**	estimated 28.8 independent-window FA/h
Corrected post-TSE hybrid	84.48%	75.38%	79.67%	larger hard-negative test
AISHELL-augmented KWS	83.15%	78.46%	80.74%	broader coverage, calibration mismatch
Runtime target KWS, negative-heavy validation	30.7%	76.9%	43.9%	1,328-row imbalanced split

The 97.40% value is an isolated KWS operating point and should not be reported as full-pipeline wake accuracy. The later open-set Smart Speaker study demonstrates why continuous negative exposure is necessary.

6.4 Stage 3b speaker verification

Table 7: Compact cabin+Smart speaker verifier. The threshold was selected on validation and applied to the held-out Eval1 cache.

Measure	Result
Dev AUC	0.9380
Eval1 AUC	**0.9706**
Validation selected threshold	0.6451
Test TPR at selected threshold	**79.69%**
Test FPR at selected threshold	**0.534%**
Test precision at selected threshold	92.73%

The standalone verifier cache reported 93.0% of synthetic target-present pairs, 2.0% of AISHELL cabin negatives, and 0.0% of Smart Speaker negatives at an earlier threshold of 0.30. These are verifier-cache metrics, not the continuous full-pipeline open-set result. The LibriSpeech-only verifier was a stronger generic mismatch rejector but lost same-speaker positives. A matched clean wake/enrollment proxy detected 15/16 wakes (93.75%) without the compact gate, 3/16 (18.75%) with compact gate 0.30 and KWS 0.50, and 15/16 (93.75%) with compact gate 0.30 and KWS 0.46. The corrected continuous Smart Speaker study still reports 1.88/h at KWS 0.50 and 154.74/h at KWS 0.46, so the compact gate is a research candidate, not a drop-in production improvement; its positive and negative calibration must be redone with same-driver cabin recordings.

6.5 Stage 4 intent classification

Table 8: Intent results for the retained wav2vec2 FSC classifier.

Evaluation	Examples	Top-1	Top-5	Macro precision	Macro F1
Official FSC test	3,793	**99.31%**	**99.89%**	99.25%	99.24%
TSE-domain validation	2,314	90.49%	98.01%	90.36%	90.13%

Runtime waveform calibration on the TSE-domain cache reported ECE 0.023, Brier score 0.135, and mean confidence 0.913. A 0.53 confidence threshold gave 94.12% coverage at 93.66% selective accuracy on validation-domain evidence; this threshold is not frozen for production.

7 End-to-end results

7.1 Positive cross-corpus proxies

Table 9: Positive continuous proxies. These are not real-driver tests: enrollment, wake, and command are not guaranteed to be the same speaker.

Proxy	Wake	Intent	Rows, conditioned result, latency, and caveat
Expanded splitgate proxy	78.1%	68.8%	32 rows; conditioned intent 88.0%; 2.1 s mean / 2.5 s p95; cross-corpus sequence.
Selected eight-scene safety proxy	100%	100%	8 scenes; conditioned intent 100%; 2.5 s; short and synthetic.
Clean endpoint trim, 500 ms hop	**100%**	**84.4%**	128 rows; conditioned intent 84.4%; latency not comparable; 64 cross-corpus wake groups.
Clean buffered hybrid, 1 s hop	87.5%	76.6%	128 rows; conditioned intent 87.5%; decoder diagnostic, latency not comparable.
Noisy CAVEMOVE transfer	45.3%	25.8%	128 rows; conditioned intent 56.9%; latency not comparable; clean Stage 1 does not transfer.
ReMASC vehicle-stress fast stream	96.9%	81.3%	448 s; conditioned intent not reported; 0.968 s mean / 1.625 s p95; diagnostic margin was negative.

The clean endpoint score exceeds the provisional 80% proxy target, but the open-set negative test below invalidates it as a safe operating point. The positive and negative results must be interpreted together.

7.2 Long target-absent cabin negative stream

AISHELL-5 Eval2 contains 18 sessions and 3.889 hours of target-absent, speech-rich four-channel audio. The fixed LibriSpeech enrollment speaker is absent. With frame presence, target KWS, temporal smoothing, and compact speaker gating, the best compact cabin-aware diagnostic recorded zero full-path events/hour. The exact zero-event 95% Poisson upper bound is 0.95 events/hour for 3.889 hours. This is a negative-only bound, not proof of perfect recall. The separate Smart Speaker compact-backend recheck below found nonzero events after correcting a batched-evaluator backend-attribution error.

Table 10: AISHELL-5 target-absent negative stream.

Gate	Full events	Full/h	95% upper bound	KWS-only/h
SpEx cosine 0.24	18	4.63	7.00	96.69
Compact GSC verifier 0.50	7	1.80	3.36	96.69
Compact GSC verifier 0.20	11	2.83	4.73	96.69
Compact cabin-aware 0.30	**0**	**0.00**	0.95	96.69
Compact cabin+Smart 0.30, 0.25 s / 8-of-8	**0**	**0.00**	0.95	71.74

7.3 Open-set non-Marvin speech

The University of Rochester Smart Speaker Command Dataset consists of continuous clean single-channel "Hey Alexa" commands from seven speakers. Every clip is negative for "Marvin". The single channel was duplicated to four channels only to satisfy the interface. It is not a real-cabin test.

Table 11: Completed full-corpus open-set study (N = files; h = hours). The hard-negative and compact rows use the corrected full-context evaluator. The clean-only row is retained as a historical pre-correction diagnostic and is not directly comparable.

Candidate	N	h	Full/h	KWS/h	Action
Clean-only KWS (historical)	2,110	3.193	1,209.38	1,289.26	historical diagnostic
Hard-negative/TSE-domain KWS	2,110	3.193	23.81	49.80	0 false actions / 68.65 s
Compact verifier, gate .30, KWS .50	2,110	3.193	**1.88**	**38.21**	2 false wakes; 0 actions / 68.65 s
Compact verifier, gate .30, KWS .46	2,110	3.193	154.74	592.95	threshold diagnostic

The corrected full hard-negative run produced 76 full-path events and 159 KWS-only events. With the actual compact verifier, gate 0.30 produced 6 full-path events and 122 KWS-only events at KWS 0.50, or 1.88/h and 38.21/h; at KWS 0.46 it produced 494 full-path events and 1,893 KWS-only events, or 154.74/h and 592.95/h. The earlier 0.00/h compact claim was a backend attribution error: the batched evaluator had not received the compact verifier and could silently use the SpEx scorer. The historical 48.55/h value included one invalid padded tail window per affected stream and is superseded. The speaker-threshold sweep remained 24.12/h even at gate 0.80 at KWS 0.46. These are open-set diagnostics, not production FAR estimates.

A full no-gate threshold replay over the same 3.193-hour negative corpus produced 926.54, 925.28, 538.13, 23.81, 0.00, and 0.00 full-path events/hour at KWS thresholds 0.35, 0.40, 0.45, 0.50, 0.55, and 0.60, respectively. The compact-backend replay produced 171.34, 154.74, 125.92, 89.90, 39.78, and 1.88/h at KWS thresholds 0.45–0.50, with gate 0.30. Neither curve is a release calibration: the corrected noisy proxy loses all target wakes at KWS 0.55, while the compact path remains too permissive at lower thresholds. This is evidence of poor score separation and requires a same-driver tuning/test split before threshold freezing.

7.4 15-minute continuous streams

The strongest public-corpus stream evidence was a single stateful 15-minute pass at 250 ms hop, with noise beds, silence, sparse random events, and 0–15 dB cabin SNR. In Stream A the speaker used for enrollment and wake was consistent, but the command was cross-corpus. It detected 12/12 wakes with 1.49 s mean latency, but two false wakes (8.0/h); one passenger "Marvin" slipped through at 2.9 dB SNR. Stream B used command-speaker enrollment and a cross-corpus wake and detected 0/12 target wakes. These streams are useful diagnostics, not a driver study.

8 Production readiness and release gates

Table 12: Current production-readiness audit.

Gate	Status	Evidence	Reason
same-driver continuity	**blocked**	none	zero real-driver sessions
passenger rejection	**blocked**	none	no real passenger/driver sessions
study size	**blocked**	none	no consented pilot corpus
capture hardware match	**blocked**	none	host exposes no matching six-input device
frozen split	**blocked**	partial	no speaker-disjoint real-driver split
threshold freeze	**blocked**	partial	thresholds selected on proxies only
reproducible release state	**blocked**	partial	dirty worktree and evaluated-state manifest only
no-command safety	**evidence present**	proxy/negative streams	not real-cabin final evidence
latency	**evidence present**	host CPU/GPU and ONNX	target Jetson not measured
resource budget	**evidence present**	footprint and host memory	target SoC budget not measured
regression	**evidence present**	tests and dual-domain checks	does not prove field reliability

The release identity is tied to a dirty worktree, a configuration hash, and checkpoint hashes in `logs/production_readiness_audit_current_2026-10-14_v4.json`. The audit correctly reports `production_ready=false`. The current results support a research prototype or pilot paper, not a production claim.

9 Reproducibility

9.1 Primary artifacts

- Dual-domain Stage 1 regression: `logs/stage1_dual_domain_regression_full_2026-10-10.json`.
- Stage 2 validation: `logs/stage2_frame_presence_dual_domain_nonwake_ft1/metrics_last.json`.
- KWS regression: `logs/stage3_kws_gsc_marvin_tse_ft2/metrics_last.json` and `logs/stage3_target_`
- Intent: `logs/stage4_intent_wav2vec_ft8_resume_tse_domain/metrics_official_test.json` and `metrics_tse_domain.json`.
- Full benchmark: `reports/system_accuracy_latency_report_2026-10-12.md`.
- Clean/open-set follow-up: `reports/clean_generalization_and_noise_transfer_2026-10-14.md`.
- Full-context Smart Speaker threshold sweep: `reports/smart_speaker_threshold_sweep_2026-10-14.md`.
- Corrected compact-gate backend audit: `reports/compact_gate_backend_correction_2026-10-14.md`.

- Current readiness: `reports/production_readiness_snapshot_current_2026-10-14_v4.md`.
- ONNX manifest: `output/onnx/retained_ckptfix4_2026-10-14/manifest.json`.

9.2 Representative commands

```
# Unit and regression tests
python -m pytest -q

# Full system benchmark on the selected configuration
python scripts/eval/eval_continuous_full_pipeline_proxy.py --config configs/
    eval_end_to_end_cavemove_target_kws_smart_speaker_compact_cabin_smart_thr030_2026-10-12.yaml --paths
    configs/paths.yaml

# Smart Speaker open-set diagnostic (corrected compact backend)
python scripts/eval/eval_smart_speaker_command_negative.py --root D:/datasets/
    smart_speaker_command_26417548/SmartSpeakerCommandDataset_RutowskiEtAl_2023 --config configs/
    eval_end_to_end_cavemove_target_kws_smart_speaker_compact_cabin_smart_thr030_2026-10-12.yaml --paths
    configs/paths.yaml --hop-seconds 0.25 --batch-size 32 --kws-thresholds 0.45 0.46 0.47 0.48 0.49 0.50
    --action-sample-files 14

# ONNX host streaming benchmark
python scripts/eval/benchmark_onnx_streaming_runner.py --onnx-manifest output/onnx/retained_ckptfix4_2026
    -10-14/manifest.json --seconds 10 --frame-ms 20 --intra-op-threads 8 --inter-op-threads 1

# Readiness audit
python scripts/reporting/audit_production_readiness.py --config configs/
    eval_end_to_end_clean_english_identity_targetkws_endpoint08_longcommand_candidate.yaml --paths
    configs/paths.yaml --output-json logs/production_readiness_audit_current_2026-10-14_v4.json --output-
    md reports/production_readiness_snapshot_current_2026-10-14_v4.md
```

9.3 Jetson validation procedure

1. Transfer `car_tse_jetson_runtime_ckptfix4_2026-10-14.zip` and the source repository to the Jetson.
2. Unzip the runtime bundle and verify `SHA256SUMS.txt`.
3. Install the JetPack-compatible ONNX Runtime package or build the supported TensorRT/CUDA execution provider. Do not assume the Windows host provider list applies to Jetson.
4. Confirm that the runtime uses the expected four microphone inputs, aligned playback reference, 16 kHz sampling, and the input shapes recorded in the ONNX manifest.
5. Run a warm-up, then at least 60 seconds of repeated four-channel input at 250 ms and 500 ms hops. Record wall time, dropped audio, p50/p95 inference time, memory, temperature, and power.
6. Repeat with intent lazy execution and with intent forced on every hop. Keep the Jetson model, JetPack, CUDA, TensorRT, ONNX Runtime version, clocks, power mode, and thread settings in the report.

The Jetson bundle is a latency-test package, not a production package. The current ONNX export is float and its provider compatibility must be checked on the target device. Quantization and TensorRT compilation should be reported as a new, separately hashed artifact.

10 Paper claims that are currently safe

1. "The proposed cascade improves target-speaker extraction on held-out synthetic cabin mixtures and preserves CAVEMOVE performance while improving an AISHELL-5 domain diagnostic."

2. "The clean FSC intent branch achieves 99.31% top-1 and 99.89% top-5 accuracy on the official test set."

3. "Presence and compact speaker verification reduce false triggers on speech-rich target-absent diagnostic streams, with the exact reduction dependent on the operating point."

4. "The ONNX composition matches the PyTorch state-machine decisions on the deterministic parity set under the documented integration tolerance."

5. "The complete system remains a research-engineering preview because same-driver continuous cabin evidence and target-device latency are missing."

11 Claims that are not currently safe

- Do not call the system production-ready.
- Do not call the full pipeline SOTA.
- Do not turn the 99.31% clean intent score into end-to-end accuracy.
- Do not call the 0.00/h negative results proof of zero false accepts; report exposure and the 95% upper bound.
- Do not describe Smart Speaker or AISHELL-5 as a real-driver wake-to-action evaluation.
- Do not claim Jetson latency until the bundle is measured on the actual Jetson model, power mode, and software stack.

12 Recommended next experiments

1. Record the speaker-consistent cabin pilot with matched microphone and playback channels.
2. Freeze speaker-disjoint tuning and test sessions before choosing thresholds.
3. Compare the learned TSE pipeline with a classic seat-zone beamforming baseline at matched false-accept rates.
4. Train the presence/verifier/KWS operating point on real continuous labels, including passenger "Marvin" and replayed/cloned speech.
5. Export and benchmark on Jetson, then quantize or TensorRT-compile the confirmed model identity.
6. Replace or distill wav2vec2 and train a causal/chunked Stage 1 only after the data contract is fixed.
7. Re-run the complete report with held-out drivers, action labels, p50/p95 end-to-end response, WER/intent accuracy, FAR/h, FRR, and passenger rejection by condition.

13 Conclusion

The project has a coherent and reproducible research prototype. Its strongest evidence is target-speaker extraction on synthetic cabin mixtures, clean intent classification, component-level latency measurement, and a corrected ONNX deployment path. Its decisive weakness is the absence of a same-driver, continuous, real-cabin evaluation and the still-high open-set

false-wake rate of permissive configurations. The Jetson transfer package enables the next latency measurement, but it does not change the evidence boundary. Production readiness requires the real-driver pilot, target hardware measurement, frozen thresholds, and a safety study with sufficient negative exposure.

A Metric definitions

SI-SDRi Improvement in scale-invariant signal-to-distortion ratio between the mixture and extracted target relative to the clean target.

Wake recall Fraction of labeled target-driver wake events that produce an accepted wake event within the allowed timing tolerance.

Intent accuracy
Fraction of labeled post-wake commands whose predicted FSC intent matches the label. Conditioned accuracy is computed only among commands after a correctly accepted wake.

FAR False-accept rate for binary presence or speaker decisions on a labeled evaluation split.

False wakes/hour
Number of merged wake events divided by negative audio exposure in hours. This is not the same as frame-level FPR.

False actions/hour
Number of vehicle actions emitted when no target command was expected, divided by the corresponding exposure.

RTF Inference wall time divided by replayed audio duration. RTF below one indicates that a runner can keep up with the replay under the measured host conditions.

B Artifact and model transfer inventory

Table 13: Files included in the two transfer archives created with this report.

Archive	Contents	Purpose
`car_tse_all_checkpoints_2026-10-14.zip`	all 95 local checkpoint files under `checkpoints/`, including experimental variants	complete PyTorch archive
`car_tse_jetson_runtime_ckptfix4_2026-10-14.zip`	retained ckptfix4 ONNX graphs, manifest, runtime config, labels, and Jetson README/checksums	practical Jetson latency transfer

The generated archive sizes are **456,858,624 bytes (435.69 MiB)** for the Jetson runtime bundle and **2,757,225,261 bytes (2.568 GiB)** for the complete checkpoint archive. Their SHA-256 digests are respectively `7503E366E6D4BF13271FB2FA3BC65493FC0F4509136CFB8BB21D7BAD85407CBC` and `34D23DFB02E6FBB0DD8824B92CEC6B5596A1887136720A0554D2704F19B9085B`. The checkpoint archive contains 95 model files; the Jetson archive contains the retained ONNX runtime plus the optional compact export and benchmark support files.

C References and citation placeholders

The final paper should replace these placeholders with the canonical bibliographic records required by the relevant licenses and venues:

- SpEx+ target-speaker extraction paper and the original SpEx paper.
- wav2vec 2.0: Baevski et al., "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations," NeurIPS 2020.
- Google Speech Commands dataset and keyword spotting baseline.
- Fluent Speech Commands dataset and spoken-language-understanding benchmark.
- LibriSpeech and Libri2Mix.
- AISHELL-5 multi-channel far-field speech corpus.
- CAVEMOVE or the exact cabin-noise/multichannel source used by the local manifest.
- University of Rochester Smart Speaker Command Dataset, DOI 10.60593/ur.d.26417548, CC BY 4.0 as recorded in the local evaluator.

D Ablation studies

D.1 Component ablation under passenger speech

The eight-scene cross-corpus matrix used two target wake clips, four cabin SNRs (-5, 0, 10, and 20 dB), 14 seconds per scene, and passenger speech at several relative levels. The target and command speakers were not verified to be the same person.

Table 14: Passenger loudness ablation. The no-TSE and no-presence variants are diagnostic variants, not recommended deployments.

Passenger level	**Full**		**No TSE**		**No presence**	
	Wake	FA/h	Wake	FA/h	Wake	FA/h
-10 dB (louder)	25%	96.4	50%	160.7	100%	450.0
0 dB	25%	0.0	50%	64.3	100%	417.9
+10 dB (quieter)	25%	0.0	50%	32.1	100%	289.3

Removing presence increases apparent wake recall while sharply increasing false actions. This identifies presence and target selectivity, rather than static cabin noise alone, as the main engineering risk.

D.2 Temporal and enrollment ablation

Table 15: Runtime temporal settings on the same eight-scene proxy.

Hop and rule	**Wake recall**	**Intent**	**FA/h**	**Mean latency**
0.5 s, one hit	100%	0%	32.1	0.875 s
0.5 s, 2-of-2	100%	0%	32.1	1.375 s
0.5 s, 2-of-3 + refractory	25%	12.5%	0	1.500 s
0.5 s, 3-of-5 + refractory	0%	0%	0	–
1.0 s, one hit	100%	0%	0	1.250 s
1.0 s, 2-of-2	25%	12.5%	0	1.500 s
1.0 s, 2-of-3 + refractory	12.5%	12.5%	0	2.500 s

The proxy does not define a safe operating point. Stricter confirmation can suppress false actions but can also suppress the target. Thresholds must be selected on a speaker-consistent tuning split.

D.3 Channel and speaker gate ablation

Replacing four channels with repeated mono, front-pair, or rear-pair inputs gave the same 25% wake and 12.5% intent on the small proxy. This does not prove that microphone geometry is irrelevant; the simulation repeats the same source rather than recording real seat zones.

The proxy's mean speaker similarity was only 0.031. A threshold of 0.0 passed 72.3% of windows with no outcome change, while 0.2 or higher rejected all windows and gave 0% wake recall. This is why the gate cannot be calibrated from the cross-corpus proxy.

D.4 Verifier scale and fusion

Table 16: Speaker-verifier data scaling and ensemble diagnostic at threshold 0.30.

Model	GSC positives	Libri positives	Libri negatives	AISHELL	Smart Speaker
Cabin+Smart v2	**93.0%**	75.6%	45.9%	2.0%	0.0%
LibriSpeech v3	68.0%	68.6%	**18.9%**	**0.0%**	0.0%
Mean ensemble	86.7%	**78.3%**	36.3%	0.5%	0.0%
Minimum ensemble	67.2%	57.6%	14.3%	0.0%	0.0%

No fusion dominates. The v2 cabin+Smart verifier retains the best synthetic positive rate and is the current research baseline; real-driver data is needed to choose the safety/recall tradeoff.

E Latency, footprint, and deployment

E.1 Measurement contract

Component latency was measured with 30 repeats and 5 warm-up iterations on an RTX 4080 GPU in FP16, and 15 repeats and 3 warm-up iterations on the host CPU in FP32. Inputs were random three-second four-channel windows, with a two-second intent input. These are model-forward measurements: microphone I/O, resampling, buffering, scheduling, thermal behavior, and vehicle action execution are excluded.

E.2 Component latency

Table 17: Per-component latency in milliseconds. Values are mean / p50 / p95.

Component	GPU FP16	CPU FP32	Parameters
Stage 1 extraction, 3 s x 4 ch	66.46 / 62.78 / 96.51	423.04 / 420.79 / 439.47	10.73 M
Enrollment speaker embedding, 3 s	4.84 / 4.85 / 4.97	23.16 / 23.96 / 24.90	10.73 M
Extracted speaker embedding, 3 s	4.90 / 4.87 / 5.05	23.40 / 23.58 / 25.00	10.73 M
Frame presence, 3 s	1.57 / 1.57 / 1.60	1.13 / 1.13 / 1.15	0.034 M
Target KWS, 1 s	2.20 / 2.20 / 2.28	2.11 / 2.02 / 2.29	0.39 M
General KWS, 1 s	1.47 / 1.45 / 1.56	1.97 / 1.95 / 2.10	0.09 M
Compact verifier, 2 x 1 s	5.04 / 5.02 / 5.17	3.04 / 2.96 / 3.39	0.18 M
Intent wav2vec2, 2 s	17.41 / 15.66 / 21.50	94.93 / 94.73 / 99.64	94.78 M

E.3 Composite latency

Table 18: Full reference pipeline model-forward timing.

Measurement	Without compact gate	With compact cabin+Smart gate
GPU FP16 wake stage, mean / p50 / p95	48.55 / 48.21 / 51.56 ms	56.99 / 56.94 / 60.84 ms
GPU FP16 post-wake command stage	62.34 / 62.57 / 64.44 ms	76.21 / 76.11 / 80.27 ms
CPU FP32 wake stage	522.22 ms	561.15 / 562.82 / 576.69 ms
CPU FP32 post-wake command stage	641.67 ms	679.57 / 677.89 / 715.99 ms
Enrollment embedding, first call	–	604.11 ms GPU; 466.25 ms CPU
Enrollment embedding, steady state	–	approx. 5 ms GPU; approx. 23 ms CPU

E.4 Streaming runner

Table 19: Streaming simulation on the development host. RTF below 1.0 means the runner keeps up with the replayed audio.

Run	Hop	RTF	Skipped audio	Peak lag
GPU capture-thread simulation	250 ms	0.28x	0.0 s / 60 s	0.2 s
CPU capture-thread simulation	250 ms	1.73x	9.0 s / 30 s	4.85 s
CPU capture-thread simulation	1.0 s	**0.47x**	0.0 s / 60 s	0.9 s

E.5 ONNX deployment path

The retained ONNX export is `output/onnx/retained_ckptfix4_2026-10-14`. It contains Stage 1, speaker embedding, presence, general KWS, target KWS, compact verifier, and intent graphs. The manifest reports all seven component parity checks as passed and includes input contracts. Stage 1 consumes `[1,4,48000]` microphone audio, `[1,24000]` enrollment, and `[1,48000]` playback reference; intent consumes `[1,32000]` audio.

Table 20: ONNX streaming optimization.

Mode	Inference wall	Processed audio	Inference RTF	Dropped
Full components every 250 ms	18.765 s	7.25 s	2.588x	0 s
Lazy intent, only after wake	16.161 s	7.25 s	2.229x	0 s
Lazy intent, 500 ms diagnostic	8.195 s	7.50 s	1.093x	0 s

Lazy intent reduced measured inference wall time by approximately 13.9%. The host CPU still does not sustain a 250 ms hop. The next valid performance step is Jetson or automotive-SoC-specific compilation and quantization, or a model trained with a causal/chunked Stage 1.

E.6 Footprint

Table 21: Retained PyTorch reference footprint.

Item	Value
Total parameters	106,203,762
Checkpoint bytes	425,242,274 bytes (425.24 MB decimal)
Intent branch	94.78 M parameters, approximately 89%
Peak CUDA allocation, compact FP16 wake benchmark	486 MB
Older FP32 CUDA peak	up to 9.9 GB, not an edge-safe default
Retained ONNX ckptfix4 directory	approximately 411.4 MiB
All local checkpoint files	approximately 3.07 GiB, including experiments